BUSINESS CASE

Project Presentation 372S

Team Leader: Oluhle Makhaye 222419636
Researcher: Entle Myezo 230076238
Writer/Editor: Mihlali Mdlangazi 222035749
Presenter: Lukhanyo Mweli 222830646
Designer: All members

Cape Peninsula University of Technology
ICT: Applications Development

WASHTRACK

Executive Summary

Students at New Market Junction in Woodstock, Cape Town, are consistently losing prepaid laundry points due to malfunctioning washing machines. With only ten points allocated per month, each lost point directly impacts students financially and operationally.

WASHTRACK is a digital platform that provides real-time machine status, booking functionality, and fault reporting to eliminate wasted points and improve operational efficiency. The solution benefits both students and property management by reducing frustration, improving maintenance response times, and increasing overall satisfaction.

This business case proposes the development and phased implementation of WASHTRACK, with potential for scaling across multiple student housing facilities.

Problem Statement

Residents at New Market Junction face a recurring issue where they unknowingly use broken laundry machines, resulting in lost points with no refund mechanism

Key issues include:

- No visibility into machine functionality before use.
- No system for reporting faults or recovering lost points.
- High time wastage due to trial-and-error usage.
- Lack of accountability from management systems.

This problem occurs daily, particularly during peak usage hours, and significantly impacts students' finances, time management, and academic focus.

Evidence & Validation

Primary research was conducted through interviews with eight residents.

Key findings:

- Average loss: **2.5** points per student per month (25% of allocation).
- Average frustration level: **4.1 / 5**
- Time wasted: **1–4** hours per week per student.
- **75%** of respondents are willing to pay for a solution.

The data confirms that:

- The problem is widespread.
- It has measurable financial and time impact.
- There is clear demand for a solution.

Proposed Solution

WASHTRACK is a mobile-enabled platform designed to provide transparency and control over laundry usage.

Core features:

- Real-time machine status (working, occupied, broken).
- Booking system for machine reservations.
- Fault reporting with timestamps and images.
- Laundry point balance tracking.
- Push notifications for bookings and cycle completion.

Future integration with South Point systems would enable:

- Automated refunds for failed machine usage.
- Real-time card balance synchronization.

Objectives

The solution aims to achieve:

- Reduce lost laundry points by at least **80%**.
- Reduce time spent searching for machines by **50%** or more.
- Improve student satisfaction levels.
- Reduce complaint volume to management.
- Improve maintenance response efficiency.

Value Proposition

For Students:

- Protect limited financial resources (laundry points).
- Save time and reduce stress.
- Improve planning and convenience.

For South Point:

- Reduced complaints and operational friction.
- Data-driven maintenance decisions.
- Improved tenant satisfaction and retention.
- Competitive advantage in student housing market.

Revenue Model

Three viable revenue streams:

1. Subscription Model
   R30–R50 per month per student
2. Pay-Per-Use Model
   R5–R10 per booking
3. Enterprise Model
   South Point pays an annual fee to provide the service as a built-in amenity.

Feasibility Analysis

Technical Feasibility

- Phase 1 web app is easily achievable using your current stack (Node.js, MongoDB, APIs).
- IoT integration is more complex but scalable in later phases.

Operational Feasibility

- Students already use smartphones, adoption barrier is low.
- Requires cooperation from South Point for full integration.

Financial Feasibility

- Low initial development cost (student-built MVP).
- High scalability across multiple residences.
- Recurring revenue potential.

Risks & Mitigation

Risk	Impact	Mitigation
Low student adoption	Medium	Simple UX + incentives
South Point rejects integration	High	Launch standalone MVP first
Incorrect machine status data	Medium	Combine user reporting + verification
Students unwilling to pay	High	Focus on enterprise model

Implementation Plan

Phase 1: MVP (0–3 months)

- Web-based mobile app
- Manual machine status updates
- Basic booking system
- Fault reporting

Phase 2: System Integration (3–6 months)

- IoT sensors for automation
- Card system integration
- Push notifications

Phase 3: Expansion (6–12 months)

- Rollout to other South Point properties.
- Standardized platform deployment.

Conclusion & Recommendation

The laundry point loss issue at New Market Junction is a validated, high-frequency problem with measurable financial and operational impact.

WASHTRACK provides a practical, scalable, and technology-driven solution that delivers value to both students and property management.

Recommendation:

- Proceed with MVP development immediately.
- Engage South Point early for partnership opportunities.
- Prioritize enterprise adoption for long-term scalability.